Atlas 148: Sharp Paw Bias and a Two-Dimensional Hilbert Projection

Abstract

We prove the chromatic-polynomial lower bound for NetworkX Atlas graph 148 on the full interval $p \in [1/2, 1]$ and for every graphon. On $[1/2, 3/5]$, a Hilbert projection reduces its density to the square of a paw density. Symmetrizing the paw leaf, biasing by $d^{1/3}$, and applying the sharp triangle-density theorem gives the required strengthened paw bound. On $[3/5, 1]$, a two-dimensional projection retains the covariance term that is lost by the false product comparison with two triangles and a 4-cycle. The remaining scalar supporting line is certified exactly on four rational Bernstein boxes.

1 The graph, kernels, and target

Let $(\Omega, \mathcal{A}, \mu)$ be a probability space and let $W\colon \Omega^2 \to [0,1]$ be symmetric and jointly measurable. Put

$$p = \int_{\Omega^2} W(x,y)\,\mathrm{d}\mu^2, \qquad d(x) = \int_{\Omega} W(x,y)\,\mathrm{d}\mu(y),$$

and define the common-neighbour and triangle-edge kernels

$$S(x,y) = \int_{\Omega} W(x,z)W(y,z)\,\mathrm{d}\mu(z), \qquad K(x,y) = W(x,y)S(x,y).$$

NetworkX Atlas 148, with graph6 code `EyUG`, has edge list

$$01, 02, 05, 12, 13, 14, 34, 45.$$

It consists of the triangles 012 and 134, which share vertex 1, together with the two-edge path 0–5–4 between the indicated outer vertices. The triangle gives chromatic number at least three, while the color classes $\{0,3\}, \{1,5\}, \{2,4\}$ give a proper three-colouring. Thus its chromatic number is three and the admissible interval is $p \in [1/2, 1]$.

With $x = 1$, $a = 0$, and $b = 4$, integrating vertices $2, 3, 5$ gives the exact identity

$$t(F_{148}, W) = \int_{\Omega^3} K(x,a)K(x,b)S(a,b)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(a)\,\mathrm{d}\mu(b). \tag{1}$$

To count proper r-colourings, first colour x, then the two outer vertices a, b, then the remaining vertex of each triangle, and finally the midpoint of the outer path. If a, b have the same colour, the last vertex has $r-1$ choices; if they have distinct colours, it has $r-2$ choices. Hence the count is $r(r-2)^2\big((r-1)^2 + (r-1)(r-2)^2\big)$, and the chromatic polynomial and target are

$$\chi_{F_{148}}(r) = r(r-1)(r-2)^2(r^2 - 3r + 3), \tag{2}$$

$$\Phi_{F_{148}}(p) = p(2p-1)^2(3p^2 - 3p + 1). \tag{3}$$

Write

$$q = 1 - p, \qquad c = 2p - 1, \qquad f = 3p^2 - 3p + 1 = p^3 + q^3.$$

2 The normalized sharp triangle profile

We use the graphon form of the clique density theorem for triangles. Define $g\colon [0,1] \to [0,1]$ as follows. Choose an integer $k \geq 1$ and $u \in [1/(k+1), 1/k]$ such that

$$s = 1 - ku^2 - (1-ku)^2, \tag{4}$$

put

$$g(s) = 1 - 3\big(ku^2 + (1-ku)^2\big) + 2\big(ku^3 + (1-ku)^3\big), \tag{5}$$

and set $g(1) = 1$. The definitions agree at the common endpoints.

Theorem 2.1 (Sharp triangle density)**.** *Every graphon V of edge density s satisfies*

$$t(K_3, V) \geq g(s).$$

This is the K_3 instance of the clique density theorem. Its usual finite asymptotic and standard-graphon forms imply the stated arbitrary-probability- space form by sampling independent points from the given space and then independent Bernoulli edges. Edge and triangle densities converge in probability to their graphon values, so no atomlessness assumption is needed.

The feature needed here is an elementary consequence of the explicit sharp profile.

Lemma 2.2 (Normalized triangle monotonicity)**.** *The function $g(s)/s^{3/2}$ is nondecreasing on $(0,1]$.*

Proof. On the branch (4), differentiate with respect to u. Direct cancellation gives

$$s^{5/2}\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{g(s)}{s^{3/2}}\right) = \frac{3}{2}k(k-1)u^2 \geq 0.$$

The profile and the normalized ratio are continuous at every branch endpoint. This proves the claim globally. For $k = 1$, both sides vanish on the triangle-free interval. Since $g \geq 0$ and $s^{3/2}$ is increasing, the lemma also shows that g itself is nondecreasing. □

3 A sharp paw lower bound

Let

$$\tau(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z), \qquad G = \int_{\Omega} d(x)\tau(x)\,\mathrm{d}\mu(x).$$

Thus G is the paw homomorphism density.

Lemma 3.1 (Fractionally degree-weighted edge)**.** *For $0 < \alpha \leq 1$,*

$$N_\alpha := \int_{\Omega^2} W(x,y)d(x)^\alpha d(y)^\alpha\,\mathrm{d}\mu^2 \geq p^{1+2\alpha}.$$

Proof. Put $\theta = 1/(1+2\alpha)$ and $\eta = \alpha/(1+2\alpha)$. On the support of W,

$$W = (Wd(x)^\alpha d(y)^\alpha)^\theta (W/d(x))^\eta (W/d(y))^\eta.$$

Hölder's inequality applies after defining the quotients to be zero on zero-degree fibres. Each quotient integral is at most one, and hence $p \leq N_\alpha^\theta$. □

Lemma 3.2 (Sharp paw bias)**.** *Every graphon of edge density $p \geq 1/2$ satisfies*

$$G \geq p\, g(p). \tag{6}$$

Proof. Average the three equal representations obtained by putting the paw leaf at each triangle vertex. Arithmetic–geometric mean gives

$$G \geq \int_{\Omega^3} W(x,y)W(x,z)W(y,z)(d(x)d(y)d(z))^{1/3}\,\mathrm{d}\mu^3.$$

Put $M = \int d^{1/3}$, $z = M^3$, and use the probability measure $\mathrm{d}\nu = d^{1/3}\,\mathrm{d}\mu/M$. The right side is

$$z\, t(K_3, W; \nu).$$

The edge density s under ν obeys, by Lemma 3.1 with $\alpha = 1/3$,

$$s \geq s_0 := p\left(\frac{p}{z}\right)^{2/3}.$$

Concavity of $d^{1/3}$ gives $z \leq p$, while $s \leq 1$ implies $s_0 \leq 1$; moreover $z \leq p$ gives $s_0 \geq p$. The sharp triangle theorem, monotonicity of g, and Lemma 2.2 yield

$$\begin{aligned} G \geq z g(s_0) &= p^{5/2}\frac{g(s_0)}{s_0^{3/2}} \\ &\geq p^{5/2}\frac{g(p)}{p^{3/2}} = p g(p). \end{aligned}$$

□

4 The low-density interval

For $p \in [1/2, 2/3]$, write the sharp triangle profile using part masses $((1-y)/2, (1-y)/2, y)$:

$$p = \frac{1 + 2y - 3y^2}{2}, \qquad g(p) = \frac{3}{2}y(1-y)^2. \tag{7}$$

The relation in (7) is increasing for $0 \leq y \leq 1/3$, and its value at $y = 1/8$ is $77/128 > 3/5$. Consequently $p \leq 3/5$ gives $0 \leq y \leq 1/8$.

Lemma 4.1 (Low-interval scalar comparison)**.** *For $p \in [1/2, 3/5]$,*

$$p\, g(p)^2 \geq c^2 f.$$

Proof. Substitution from (7) gives

$$p g(p)^2 - c^2 f = \frac{y^2}{8} R(y),$$

where

$$R(y) = 1 + 6y - 159y^2 + 756y^3 - 1521y^4 + 1422y^5 - 513y^6.$$

After $y = u/8$, its degree-six Bernstein coefficients on $0 \leq u \leq 1$ are

$$1,\ \frac{9}{8},\ \frac{347}{320},\ \frac{2437}{2560}, \frac{15909}{20480},\ \frac{19349}{32768}, \frac{108079}{262144}.$$

They are all positive. Since the Bernstein basis is nonnegative on $[0,1]$, this proves $R(y) > 0$ throughout the required interval. □

For fixed x, put $F_x(a) = K(x,a)$. Since S is the kernel of T_W^2, equation (1) gives

$$t(F_{148}, W) = \int_\Omega \|T_W F_x\|_2^2 \,\mathrm{d}\mu(x). \tag{8}$$

Projection onto the constant function, followed by Jensen, gives

$$t(F_{148}, W) \geq \int_\Omega \left(\int_\Omega d(a) K(x,a) \,\mathrm{d}\mu(a) \right)^2 \mathrm{d}\mu(x) \geq G^2. \tag{9}$$

Lemma 3.2 and Lemma 4.1 therefore prove

$$t(F_{148}, W) \geq p^2 g(p)^2 \geq pc^2 f$$

throughout $p \in [1/2, 3/5]$.

5 The two-dimensional projection

For each x, put

$$s(x) = \int W(x,a)^2 \,\mathrm{d}\mu(a), \qquad v(x) = s(x) - d(x)^2,$$
$$g(x) = \int d(a) K(x,a) \,\mathrm{d}\mu(a), \qquad h(x) = \int W(x,a) S(x,a)^2 \,\mathrm{d}\mu(a).$$

The functions $\mathbf{1}$ and $W(x,\cdot) - d(x)\mathbf{1}$ are orthogonal, with squared norms 1 and $v(x)$. Their inner products with $T_W F_x$ are $g(x)$ and $h(x) - d(x)g(x)$. Bessel's inequality gives

$$\|T_W F_x\|_2^2 \geq g(x)^2 + \frac{(h(x) - d(x)g(x))^2}{v(x)}, \tag{10}$$

where the quotient is zero when $v(x) = 0$; its numerator then also vanishes.

Set

$$D = \int_{\Omega^2} W(x,y) S(x,y)^2 \,\mathrm{d}\mu^2,$$
$$L = \int_{\Omega^2} W(x,y) S(x,y) d(x) d(y) \,\mathrm{d}\mu^2,$$
$$\Delta = D - L, \qquad V = \int_\Omega v(x) \,\mathrm{d}\mu(x).$$

Then $\int g = G$ and $\int (h - dg) = \Delta$. Two applications of Cauchy–Schwarz in (10) give

$$t(F_{148}, W) \geq G^2 + \frac{\Delta^2}{V}. \tag{11}$$

Here and below the quotient is defined to be zero when $V = 0$; in that case $\Delta = 0$ by the preceding fibrewise observation. Moreover

$$V = \int W^2 - \int d^2 \leq p - p^2 = pq.$$

Thus, for $p < 1$,

$$t(F_{148}, W) \geq G^2 + \frac{\Delta^2}{pq}. \tag{12}$$

6 The high-density supporting line

Lemma 6.1 (High-density linear estimate)**.** *If $p \in [3/5, 1]$, then*

$$p^2 G - q\Delta \geq pcf. \tag{13}$$

Proof. Symmetry and arithmetic–geometric mean give

$$G = \int WS\frac{d(x)+d(y)}{2} \geq \int WSZ, \qquad Z = \sqrt{d(x)d(y)}.$$

Also $L = \int WSZ^2$ and $D = \int WS^2$. Hence the left side of (13) is at least

$$\int_{\Omega^2} W(x,y)F_p(Z,S)\,\mathrm{d}\mu^2, \qquad F_p(z,s) = s(p^2 z + qz^2 - qs). \tag{14}$$

Cauchy–Schwarz and the product union bound give

$$(2z-1)_+ \leq S \leq z.$$

For fixed z, $F_p(z,s)$ is concave in s, so it is bounded below by the smaller of its endpoint values

$$\begin{aligned} A_p(z) &= (2z-1)_+\big(p^2 z + qz^2 - q(2z-1)_+\big),\\ B_p(z) &= z^2(p^2 - q + qz). \end{aligned}$$

For $z \geq 1/2$, put

$$\ell_p(z) = cf + m_p(z-p), \qquad m_p = -2p^3 + 15p^2 - 14p + 4.$$

On $p \in [3/5, 1]$, one has $m_p \geq 0$: after substituting $p = 3/5 + 2a/5$, its degree-three Bernstein coefficients on $0 \leq a \leq 1$ are

$$\frac{71}{125}, \quad \frac{61}{75}, \quad \frac{5}{3}, \quad 3.$$

Also

$$\ell_p(1/2) = \frac{c^2}{2}(p^2 - 4p + 2) \leq 0.$$

Furthermore

$$A_p(z) - \ell_p(z) = -(z-p)^2\big(2p^2 + 2pz - 9p - 2z + 5\big) \geq 0,$$

because the bracket is decreasing in z and at $z = 1/2$ equals $2(p^2 - 4p + 2) \leq 0$. Indeed, $p^2 - 4p + 2$ is decreasing on this interval and has value $-1/25$ at $p = 3/5$.

The remaining polynomial $B_p(z) - \ell_p(z)$ is nonnegative on the full rectangle $3/5 \leq p \leq 1$, $0 \leq z \leq 1$. For completeness, substitute $p = 3/5 + 2a/5$ and affinely map each of the four z-intervals

$$[0, 1/2], \quad [1/2, 2/3], \quad [2/3, 3/4], \quad [3/4, 1]$$

to $0 \leq b \leq 1$. Its native bidegree is $(4, 3)$, and its four exact Bernstein coefficient matrices are

$$\begin{pmatrix} 178/625 & 713/3750 & 691/7500 & 51/1250 \\ 48/125 & 97/375 & 89/600 & 181/2000 \\ 2/3 & 23/50 & 13/45 & 107/600 \\ 6/5 & 13/15 & 71/120 & 31/80 \\ 2 & 3/2 & 13/12 & 3/4 \end{pmatrix},$$

$$\begin{pmatrix} 51/1250 & 533/22500 & 397/33750 & 116/16875 \\ 181/2000 & 641/9000 & 13/225 & 58/1125 \\ 107/600 & 191/1350 & 451/4050 & 4/45 \\ 31/80 & 23/72 & 7/27 & 28/135 \\ 3/4 & 23/36 & 29/54 & 4/9 \end{pmatrix},$$

$$\begin{pmatrix} 116/16875 & 299/67500 & 337/90000 & 101/20000 \\ 58/1125 & 109/2250 & 1697/36000 & 153/3200 \\ 4/45 & 629/8100 & 41/600 & 293/4800 \\ 28/135 & 49/270 & 227/1440 & 87/640 \\ 4/9 & 43/108 & 17/48 & 5/16 \end{pmatrix},$$

$$\begin{pmatrix} 101/20000 & 269/30000 & 77/2500 & 48/625 \\ 153/3200 & 299/6000 & 209/3000 & 14/125 \\ 293/4800 & 47/1200 & 8/225 & 4/75 \\ 87/640 & 17/240 & 1/40 & 0 \\ 5/16 & 3/16 & 1/12 & 0 \end{pmatrix}.$$

Every entry is nonnegative. The tensor-product Bernstein basis is nonnegative and sums to one on each box, so these matrices prove $B_p(z) \geq \ell_p(z)$ on the full rectangle. Thus $F_p(z,s) \geq \ell_p(z)$ also when $z < 1/2$, since $A_p(z) = 0 \geq \ell_p(z)$ there.

Finally use the edge-biased probability measure $W(x,y)\,\mathrm{d}\mu^2/p$. The edge geometric-mean inequality gives

$$\mathbb{E}\sqrt{d(x)d(y)} \geq p.$$

Indeed, after defining zero-degree quotients to be zero, Cauchy–Schwarz applied to $\sqrt{W/d(x)}$ and $\sqrt{W/d(y)}$ gives $\int W/Z \leq 1$. A second application of Cauchy–Schwarz gives $p^2 \leq (\int WZ)(\int W/Z)$. Since $m_p \geq 0$, integration of the supporting line proves

$$\int W F_p(Z,S) \geq p\ell_p(\mathbb{E}Z) \geq pcf.$$

□

Lemma 6.2 (Projection closes the high interval)**.** *For $p \in [3/5, 1]$,*

$$t(F_{148}, W) \geq pc^2 f.$$

Proof. The case $p = 1$ is immediate. For $p < 1$, put

$$G_0 = p^2 c, \qquad \Delta_0 = -pcq^2.$$

Then

$$G_0^2 + \frac{\Delta_0^2}{pq} = pc^2(p^3 + q^3) = pc^2 f.$$

Lemma 6.1 is precisely

$$GG_0 + \frac{\Delta\Delta_0}{pq} \geq pc^2 f.$$

Cauchy–Schwarz in this weighted two-dimensional Euclidean space and (12) finish the proof. □

7 Conclusion and scope

Theorem 7.1 (Atlas 148, graph6 `EyUG`). *Every graphon W of edge density $p \in [1/2, 1]$ satisfies*

$$\boxed{t(F_{148}, W) \geq p(2p-1)^2(3p^2-3p+1) = \Phi_{F_{148}}(p).}$$

The low- and high-density arguments meet at $p = 3/5$. At $p = 1/2$ the target is zero (with equality on the balanced complete bipartite graphon), and at $p = 1$ both sides are one. No classification of all interior equality cases is asserted. All integral identities use bounded nonnegative measurable functions, except for ordinary Hilbert-space inner products of bounded kernels, so Tonelli–Fubini applies on an arbitrary probability space. Zero-degree and zero-variance fibres were handled before division.

This proof does not use the false comparison $p^2 t(F_{148}) \geq t(K_3)^2 t(C_4)$. The two-coordinate projection retains exactly the covariance term that comparison discards. The use of the sharp triangle-density theorem is confined to the paw estimate on $[1/2, 3/5]$; the high interval uses arithmetic–geometric mean, Cauchy–Schwarz, and the displayed rational Bernstein certificate.

The Atlas identification, chromatic target, normalized-profile derivative, low-interval factorization, projection identities, and all Bernstein coefficients were reconstructed in exact rational arithmetic.

No Lean file is changed. Any future formalization would need either the graphon triangle-density theorem as an imported analytic input or a separate formal proof of its K_3 case, as well as the two-dimensional Bessel and Bernstein steps above.

References

[1] C. Reiher, *The clique density theorem*, Annals of Mathematics **184** (2016), 683–707; `https://annals.math.princeton.edu/2016/184-3/p01`.

A What the Lean formalization actually proves

The theorem above is formalized in full, on the whole interval $p \in [1/2, 1]$. The development is `lean/Taeyoung/Methods/Atlas148/`, the catalogue theorem is `satisfiesLowerBound_148`, and every declaration depends on exactly `[propext, Classical.choice, Quot.sound]`. This supersedes the closing remark of Section 7: no Lean file was left unchanged, and the two steps flagged there as obstacles turned out not to be needed in the stated form. This appendix says where the formalization departs from the argument above.

A.1 Only Fisher's band of the clique density theorem is used

Section 2 imports the full clique density theorem, and the paw estimate applies it at the tilted edge density s, which ranges over all of $[p, 1]$. Only the $k = 2$ branch is needed. Split at $s = 2/3$:

- for $s \leq 2/3$ the sharp profile is Fisher's 1989 theorem, whose parameter $c = (1 - \sqrt{4 - 6s})/3$ is exactly the y of (7);
- for $s \geq 2/3$ the crude Goodman bound $t(K_3) \geq s(2s-1)$ already suffices. Fed through the same substitution $z = p^{5/2} s^{-3/2}$ it gives $p^{5/2} s^{-1/2}(2s-1)$, increasing in s, so its minimum over $[2/3, 1]$ is at $s = 2/3$; and there it clears the requirement $p\, g(p)$ by a factor of at least 1.34 on $[1/2, 3/5]$.

The two agree at $s = 2/3$, where both equal $2/9$, so no case is lost. The Goodman branch reduces to $6g(p)^2 \le p^3$, i.e. $108y^2(1-y) \le (1+3y)^3$, whose difference factors as $(3y-1)^2(15y+1)$.

Fisher's theorem is nevertheless unavoidable: the triangle input must be within 3.5%–6% of sharp, the required ratio $c\sqrt{f/p}\,/\,g(p)$ peaking at 0.9656 at $p = 3/5$ and equalling 0.9428 at $p = 1/2$, whereas Goodman alone runs 0.667–0.848 of sharp. The Lean proof imports it from `lean/Taeyoung/Fisher/`, a copy of the sibling project `discussions/goodman-style-bound/fisher_lean` restricted to the closure of its target; `Methods/TriangleDensity.lean` restates it for this project's bundled graphons.

A.2 The fractional edge lemma needs no three-factor Hölder

Lemma 3.1 is proved above by Hölder at three factors, with the unbounded quotients $W/d(x)$. One *two*-factor Cauchy–Schwarz does it. With $u = d^{1/3}$,

$$p^2 = \Big(\iint W\Big)^2 \le \Big(\iint \frac{W}{u_x u_y}\Big)\Big(\iint W u_x u_y\Big) \le M \cdot N_{1/3},$$

the first factor being at most $M = \int d^{1/3}$ by arithmetic–geometric mean and Fubini. Combined with $M \le p^{1/3}$ — Jensen for the concave cube root — this gives $p^2 \le p^{1/3} N_{1/3}$, hence $p^5 \le N_{1/3}^3$.

Unboundedness is handled by shifting the degree to $d + \delta^3$, so that $(d+\delta^3)^{1/3} \le d^{1/3} + \delta$, and letting $\delta \to 0$; the same device gives the edge geometric mean $\iint W\sqrt{d(x)d(y)} \ge p^2$ of Section 6, which the note proves through the unbounded $\iint W/Z$.

Everything downstream is kept polynomial: the tilt sends $z = M^3$ and $s = N_{1/3}/M^2$, so the hypothesis $p^5 \le z^2 s^3$ that the scalar lemmas consume is *identically* $p^5 \le N_{1/3}^3$, the M's cancelling.

A.3 Bessel without division, and the final Cauchy–Schwarz for free

The quotient form (10) is never used. Keeping the multiplier free, for every real λ,

$$\|T_W F_x\|^2 \ge g(x)^2 + 2\lambda\big(h(x) - d(x)g(x)\big) - \lambda^2 v(x),$$

which needs neither a division nor the separate treatment of the degenerate fibres $v(x) = 0$. In that form Bessel is not a Hilbert-space statement at all: putting $w = T_W F_x - \lambda(W(x,\cdot) - d(x))$, the centred function has mean zero, so $\int w = g(x)$, and plain Jensen $(\int w)^2 \le \int w^2$ is the whole content.

The same free λ removes the two-dimensional Cauchy–Schwarz of Section 7. Integrating gives $T \ge G^2 + 2\lambda\Delta - \lambda^2 V$ with $V \le pq$; taking $\lambda = -cq$ and inserting the linear estimate (13) turns the target into

$$T - pc^2 f \ge (G - cp^2)^2 \ge 0,$$

because $f - q^3 = p^3$. So the vector (G_0, Δ_0), the weighted inner product, and the division by V all disappear; $V \le pq$ survives only as a coefficient bound.

A.4 Less Bernstein data, and no permuted triple integral

The note certifies three scalar facts with 91 Bernstein coefficients: seven for Lemma 4.1, four for the slope $m_p \ge 0$, and eighty for $B_p \ge \ell_p$ on four z-boxes. The formalization uses 40.

- Lemma 4.1 and the normalized monotonicity of Lemma 2.2 are closed directly by `nlinarith` in the coordinate y, the latter after cross-multiplying and squaring into a two-variable polynomial inequality on $0 \le y_1 \le y_2 \le 1/3$. The normalized-profile derivative also collapses: on the $k = 2$ branch, $g(s)^2/s^3 = 18y^2(1-y)/(1+3y)^3$, with derivative $36y(1-3y)/(1+3y)^4$.

- For $B_p \geq \ell_p$ the subdivision is genuinely necessary — a single box fails at every bidegree up to $(10, 12)$, because the Bernstein coefficients converge only like $O(1/n)$ along the edge $p = 3/5$, where the function's minimum is a shallow 0.0044; and a linear-programming search over all products of the four box constraints, with $(z - p)^2$ multiples adjoined, is infeasible. But *two* boxes suffice, split at $z = 7/10$ and kept at the native bidegree $(4, 3)$. Each is discharged by `linarith` against the twenty products $a^i(1 - a)^{4-i}b^j(1 - b)^{3-j}$, so no certificate datatype is involved.

The leaf symmetrization of Lemma 3.2 needs no permutation of a triple integral. Integrating the third vertex first turns each of the three degree weights into a page operator $H_s(x, y) = \int W(x, z)W(y, z)d(z)^s$, so arithmetic–geometric mean becomes the pointwise estimate

$$d(x)^{1/3}d(y)^{1/3}H_{1/3}(x, y) \leq \frac{d(x) + d(y)}{3}H_0(x, y) + \frac{H_1(x, y)}{3},$$

and one monotone step over the spine pair finishes it; the three terms on the right are all G, two by symmetry of $W \cdot H_0$ and the third because $\iint WH_1$ and G both collapse to $\int d\tau$.

A.5 Two things that are load-bearing

- **The collapse to $Z = \sqrt{d(x)d(y)}$.** Keeping $d(x)$ and $d(y)$ separate and replacing Z by $(d(x) + d(y))/2$ in the supporting line makes it *false*: it is violated by -0.014 at $p = 3/5$, $d(x) = 0.28$, $d(y) = 1$. What forces Z is that $L = \iint WSZ^2$ holds exactly, whereas $d(x)d(y) \leq \bar{d}^2$ points the wrong way.

- **The split point.** The high-density argument needs $\ell_p(1/2) \leq 0$, that is $p^2 - 4p + 2 \leq 0$, so it cannot reach below $p = 2 - \sqrt{2} \approx 0.5858$. The window $[1/2, 2 - \sqrt{2}]$ is irreducibly Fisher-dependent; a supporting line valid down to $p = 1/2$ would remove the triangle-density input from the catalogue entirely.

A.6 What is checked

The Lean graph is labelled for peeling — centre 0, the two joined outer vertices 1 and 2, the triangle apexes 3 and 4, the path middle 5 — and `Examples/Graph148.lean` transports the bound to the Atlas edge list along an explicit relabelling whose adjacency correspondence is checked by kernel computation. The chromatic polynomial (2) is proved, not assumed: its last factor $r^2 - 3r + 3$ is irreducible over the rationals, so no clique-attachment tower produces it, and the six surjective colouring counts $0, 0, 0, 18, 264, 840, 720$ are checked by kernel reduction instead, the top one through the identity $\mathrm{surj}(n) = n!$.